Game Developer’s Journal: Kaudi

Project: Kaudi – Cowrie Shell Scoring Game & Shared Collection Dice Engine
Studio: Centurion University Heritage Game Studio (CUTM)
Developer / Designer: Lead Game Engineer & Team
Master Anthology Reference: *An Anthology of Native & Indigenous Sports of India* (CUTM, 2025)

Developer's Journal Manifesto

> *"A good game requires a sound strategy for success. The Game Developer's Journal serves as physical, written evidence of strategic thinking, hard work, personal creativity, and technical prowess completed during development."*

This journal logs the day-to-day creative decisions, mathematical models, technical breakthroughs, and design challenges resolved throughout the creation of **Kaudi (Title 18)**.

Journal Entry 1: Project Genesis & Cultural Grounding

Date: September 16, 2026
Phase: Concept & Research
Focus: *Unlocking the Anthology & Deconstructing the Ancient Cowrie*

Today our studio convened to review the Centurion University *Anthology of Native & Indigenous Sports of India*. While the master collection features elaborate tactical race games like *Ashtapada*, *Chauka Bara*, and *Pachisi*, every single one of those games shares a fundamental ancestral instrument: the **Kaudi (cowrie shell)**.

Instead of treating cowries as merely a small secondary UI button in larger race games, we realized that Kaudi deserved to be its own dedicated, standalone title. Historically, village elders and merchants played quick scoring contests with handfuls of shells on temple steps. Furthermore, if we build a world-class, reusable Cowrie Dice service here first, all subsequent race games in the collection will inherit a proven, authentic dice core.

Key Decisions:

- Adopt the 6-shell and 4-shell historical configurations as selectable modes.
- Score based on mouth-up apertures, with all-up (*Chamma*) and all-down (*Ashtam*) acting as high-value critical rolls that grant extra throws.
- Committed to avoiding modern plastic dice aesthetics—the shells must look and feel like organic marine porcelain.

Journal Entry 2: Mathematical Modeling & Binomial Distributions

Date: September 18, 2026
Phase: Mathematics & Rules Engine
Focus: *Simulating Real-World Shell Physics via Discrete Probability*

Unlike a symmetrical six-sided cube where each face has a flat 1/6 (16.67%) probability, a real cowrie shell is weighted asymmetrically. The rounded dorsal back is heavier, meaning an untethered shell has approximately a 58% to 62% tendency to settle aperture-up on soft cloth, and slightly higher on hard stone.

We wrote a Python Monte Carlo script to simulate 1,000,000 throws of 6 shells assuming $P(\text{mouth up}) = 0.60$:

- 0 mouths up (*Ashtam*): ≈ 0.41% (Rare jackpot!)
- 1 mouth up: ≈ 3.68%
- 2 mouths up: ≈ 13.82%
- 3 mouths up: ≈ 27.65%
- 4 mouths up: ≈ 31.10%
- 5 mouths up: ≈ 18.66%
- 6 mouths up (*Chamma*): ≈ 4.67% (Master roll!)

This mathematical balance is extraordinary. Standard rolls consistently award 3 or 4 points, making a race to 100 points take roughly 15 to 25 turns without bonus chaining. We encoded this distribution directly into `cowrie-dice.js`, ensuring deterministic results under seedable random generators.

Journal Entry 3: Designing the "Press-Your-Luck" Risk Mechanic

Date: September 20, 2026
Phase: Game Systems & Mechanics
Focus: *Injecting Psychological Agency into Pure Probability*

During early playtesting of the Classic Target mode, we noticed that while players enjoyed the rolling animation, purely rolling and recording numbers could feel passive over long sessions.

To solve this, we introduced the **Press-Your-Luck** mode:

- Points rolled in a turn enter a temporary "Round Bank".
- The player can choose to **Bank** (securing those points permanently) or **Roll Again** to keep pushing their round total higher.
- However, if the player rolls a "Bust" (e.g., rolling only 1 mouth up on an aggressive push), their unbanked points from that turn vanish, and the turn immediately passes to their opponent.

The transformation during team playtests was instantaneous. Shouts of excitement filled the lab when a developer sat on 18 unbanked points, gambled on one more roll, hit a *Chamma*, and shot ahead to 42 points in a single turn. The psychological tension between greed and prudence became our primary gameplay hook.

Journal Entry 4: Art Direction & Centurion University Design Bible Compliance

Date: September 22, 2026
Phase: Art & User Interface
Focus: *Hand-Crafted Heritage Textures & Strict Palette Rules*

Today we received the final `00_DESIGN_BIBLE_v1.0.docx` issued by Abhi Mitra. The mandate is clear: *"Heritage, not history lesson. Readable before beautiful. Spacing always a multiple of 8px. Minimum touch target 88x88px."*

We refactored our visual styles:

- Replaced generic CSS colors with the locked design tokens:

    - Tray background: Sandstone surface `#FBF4E8` with deep carved teakwood borders `#5A3919`.
    - Deep charcoal ink typography `#17120E` ensuring a pristine 17:1 contrast ratio.
- Integrated Google Fonts: **Tiro Devanagari Hindi** for the ornate header displays and **Mukta** for crystal-clear numeric readouts.
- Ensured every player has a distinct symbolic glyph (Lotus, Elephant, Peacock, Sun Chakra) alongside color coding so the game is 100% accessible to color-blind players.

Journal Entry 5: Acoustic Foley & Low-Latency Audio Synthesis

Date: September 24, 2026
Phase: Audio Engineering
Focus: *The Acoustic Signature of Shaken Shells*

Audio makes or breaks a tactile dice game. In real life, the sound of shaking cowries in cupped hands and hearing them clatter across a wooden tray is an iconic sensory memory for anyone who grew up playing traditional games in India.

We implemented a dual-layer audio architecture in `audio.js`:

1. **Procedural Synthesis Layer**: Uses the native Web Audio API oscillators and band-pass noise filters to generate immediate, zero-latency physical impacts, woodblock clicks, and celebratory chimes. This guarantees crisp audio even if network bandwidth is limited.

2. **Recorded Foley Layer**: Multi-sampled real shell impacts that randomly modulate in pitch (± 5%) and playback volume so that no two shell collisions sound artificially identical.

3. Added an automatic gesture unlock listener on the first user interaction to bypass aggressive browser autoplay blocking policies on iOS Safari and Chrome Android.

Journal Entry 6: AI Opponent "Chanakya" & Heuristic EV Policy

Date: September 26, 2026
Phase: AI & Algorithmic Solvers
Focus: *Teaching the Computer When to Gamble*

For single-player mode, we built our AI solver, dubbed "Chanakya". Because cowrie throws are chance-driven, an adversarial tree search (like Minimax) is unnecessary. Instead, we designed an **Expected Value (EV) Decision Policy**:

$$\text{EV}(\text{Roll}) = P(\text{Safe}) \times (\text{Bank} + \text{Expected Gain}) - P(\text{Bust}) \times \text{Bank}$$

We tuned three distinct difficulty profiles:

- **Easy (Gramin)**: The AI is risk-averse. It banks as soon as its round total reaches 8–10 points, making it predictable and beginner-friendly.
- **Medium (Vyapari)**: The AI plays pure EV mathematics, banking at around 14–16 points.
- **Hard (Raja)**: The AI incorporates race context. If the human player is leading by 25 points, the AI lowers its fear threshold, taking aggressive gambles on bonus rolls to close the gap.

In automated test runs of 5,000 matches, Hard AI won 62% of games against Medium AI, proving that intelligent risk-taking yields a tangible strategic advantage.

Journal Entry 7: Final Polish, Cross-Platform QA & Production Readiness

Date: September 29, 2026
Phase: Quality Assurance & Deployment
Focus: *Zero Soft-Locks, Mobile Viewport Perfection, and Master Documentation*

We completed our final QA regression matrix today across Chrome, Safari, Edge, Firefox, and Android WebView:

- **Touch Testing**: Confirmed all interactive buttons exceed the 88px minimum touch target, with zero accidental double-tap zooming on mobile viewports.
- **Edge Case Handling**: Verified that bonus rolls cap strictly at 3 consecutive rolls, preventing edge-case runaway loops.

Kaudi is robust, culturally grounded, and ready for publication on the Centurion University Heritage Games Portal and global app stores.